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$^{+4}$Notes of Group's Theory

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** Abstrasct muito abstract . **

Notation ◇ Definition $(-)^{\diamond}$ Open $(-)^{\blacklozenge}$ Closed
□ Lemma ☐ Proposition ■ Theorem ⊠ Corollary

§ 1 Groups and Homomorphisms

Permutations

◇ Let $S_X := \{\alpha : X \to X \mid \alpha \text{ is and bijection}\}$ the set of all permutations of X and $* : S_X \times S_X \to S_X$ such that $\alpha * \beta := \alpha \circ \beta$. Usual notation:

$$X := \{1, 2, \ldots, n\} \text{ and } \alpha = \begin{pmatrix} 1 & 2 & \cdots & n \\ \alpha(1) & \alpha(2) & \cdots & \alpha(n) \end{pmatrix}.$$

◇ We call $\alpha \in S_n$ by *cycle* of length r if α fixes $n - r$ positions and $\alpha : i_1 \mapsto i_2 \mapsto \cdots \mapsto i_r \mapsto i_1$, in this case we write, $\alpha = \begin{pmatrix} i_1 & i_2 & \ldots & i_r \end{pmatrix}$. Every 1-cycle is the identity, $\mathbb{1}_X$, and 2-cycles are called by *transpositions*.

Example $S_3 \ni \alpha\beta = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (1\ \ 3)(1\ \ 3\ \ 2) = (1)(2\ \ 3).$

◇ Two cycles $\alpha, \beta \in S_X$ are *disjoint* when they affect different elements. In other words, $\{x \in X : \alpha(x) \neq x \text{ and } \beta(x) \neq x\} = \varnothing$.

■ **1.1.** Every $\alpha \in S_n$ is either a cycle or a product of disjoint cycles.

The proof is by induction on k, points moved by α. Given α, let i_1 moved by α, construct a cycle $\sigma = (i_1 \;\; i_2 \;\; \dots \;\; i_r)$, conclude that $\alpha = \sigma\alpha'$, with σ and α' disjoint, how α' moved fewer positions than α, the result follows.

◇ A *complete factorization* of α is an expresion of α as a product of disjoint cycles that includes 1-cycles for every j fixed by α.

■ **1.2.** Let $\alpha \in S_n$. A complete factorization $\alpha = \beta_1\beta_2 \dots \beta_k$ is unique except for the order in which the factors occur.

Suppose $\alpha = \gamma_1\gamma_2 \dots \gamma_s$ a second complete factorization. If β_t moves i_1, then $\exists\gamma_j$ that moves i_1, let $j = s$ (Ex. 8). Now, $\forall k \in \mathbb{N}$, $\beta_t^k(i_1) = \alpha^k(i_1) = \gamma_s^k(i_1)$ (Ex. 14), so $\beta_t = \gamma_s$ and $\beta_1 \dots \beta_{s-1} = \gamma_1 \dots \gamma_{t-1}$ (Ex. 6), the result follows by induction on $\max\{s, t\}$.

■ **1.3.** Every $\alpha \in S_n$ is a product of transpositions.

By ■ 1.1. it suffices to factor cycles: $(1 \;\; 2 \;\; \dots \;\; r) = (1 \;\; r)(1 \;\; r-1) \cdots (1 \;\; 2)$.

◇ For a permutation $S_n \ni \alpha = \beta_1 \dots \beta_t$ (complete factorization), the *signum* of α is $\operatorname{sgn}(\alpha) := (-1)^{n-t}$. *Even* or *odd* depends on the quantities of transpositions in the decomposition of α (in traspositions).

■ **1.7.** For all $\alpha, \beta \in S_n$, $\operatorname{sgn}(\alpha\beta) = \operatorname{sgn}(\alpha)\operatorname{sgn}(\beta)$. Also,

i. α is *even* $\leftrightharpoons \operatorname{sgn}(\alpha) = 1$.

ii. α is *odd* $\leftrightharpoons$ it's product of odd transpositions.

→ Doesn't support summaries, look if you want [1, pag. 9]

Groups

◇ A *group* $(G, *)$ is a nonemptyset G, equipped with an associative binary operation $* : G \times G \to G$ such that:

G1. $\exists e \in G : \forall a \in G,\ a * e = e * a = a$. → Identity element

G2. $\forall a \in G,\ \exists b \in G :\ a * b = b * a = e$. → Existence of inverses

If, in addition, the operation $*$ is commutative, then G is called an *abelian group*.

Examples • $(S_n, \circ)$ the *symetric group*. | • $(\mathbb{F}, +)$, usual addition in fields.
• $(\mathbb{Z}_n, +)$, where $\mathbb{Z}_n := \{[a] \in \mathbb{Z}/\sim : a \sim b \leftrightharpoons a \equiv b \pmod{n}\}$ • $(\mathrm{GL}_n(\mathbb{F}), \cdot)$

■ **1.10.** If G is a group, the identity and the inverse of each element are unique. $\rightarrow$ It's simply, suppose another verifying the propertie as always

⊠ **1.11.** We denote the inverse of a by a^{-1}. It follows inmediately that $(a^{-1})^{-1} = a$.

◇ Given $a \in G,\ n \in \mathbb{N}$, we define $a^n := \overbrace{a * a * \cdots * a}^{\text{n times}}$. We also set $a^0 := e$ and $a^{-n} := (a^{-1})^n$, all this expresions are *powers* of a.

Homomorphisms

◇ Let $(G, *)$ and $(H, \circ)$ be groups. A function $f : G \to H$ is a *homomorphism* if $\forall a, b \in G$, $f(a * b) = f(a) \circ f(b)$; If f is a bijection, $f : G \cong H$ (*isomorphism*).

■ **1.13.** Let $f : (G, *) \to (G', \circ)$ homomorphism. Then,

i. $f(e) = e'$. ii. $f(a^{-1}) = f(a)^{-1}$. iii. $f(a^n) = f(a)^n$.

For (i) and (ii) apply f to the equations $e = e * e$ and $a * a^{-1} = e = a^{-1} * a$. The affirmation (iii) follows by induction in n and (ii).

§ 2 Isomorphism Theorems

Subgroups

◇ A nonemptyset $S \subset G$ is a *subgroup* of G if (i) $s \in S$ implies $s^{-1} \in S$ and (ii) $s, t \in G$ imply $st \in G$. We denote $S \leq G$.

■ **2.1.** $(S, *) \leq (G, *)$ is a group in its own right. $\rightarrow *|_{S \times S}$ + definition

■ **2.2.** $S \leq G, \leftrightharpoons \mathbb{1} \in S$ and $s, t \in S$ imply $st^{-1} \in S$.

◇ Given $a \in G,\ \{a^m : m \in \mathbb{Z}\} =: \langle a \rangle \leq G$ is the *cyclic subgroup generated by* a; G is *cyclic* if $\exists a : G = \langle a \rangle$ and $|\langle a \rangle|$ is the *order* of a.

■ **2.3.** If $|\langle a \rangle| = m < \infty$, then $m = \min\{k : a^k = \mathbb{1},\ k \in \mathbb{N}\}$.

It's clear that $\exists k \in \mathbb{N} : a^k = \mathbb{1}$. Let $\{a^j\}_{j=0}^{k-1} : \forall i < l \leq k-1,\ a^i \neq a^l$, prove that $a^k = \mathbb{1}$ (by contradiction). Finally prove $\langle a \rangle = \{a^j\}$ using the division algorithm.

⊠ **2.4.** If $|G| < \infty$, then $S \leq G \ \leftrightharpoons\ s, t \in S$ imply $st \in S$. ($\Leftarrow$) If $s \in S$, then $\langle s \rangle \leq S$, $\exists m \in \mathbb{N} : s^m = \mathbb{1}$ and $s^{-1} = s^{m-1} \in S$

Note. Clearly $\{\mathbb{1}\} < H < G \leq G$, in this case H is a *proper subgroup*.

Kernel and Image of $f : G \to H$ homomorphism

- $\{a \in G : f(a) = \mathbb{1}_H\} =: \ker f \leq G$;
- $\{h \in H : h = f(a),\ a \in G\} =: \operatorname{Im} f \leq H$.

■ **2.5.** Let $(S_i)_{i \in I} : S_i \leq G$, then $\bigcap S_i \leq G$. $\to$ Mogollas

◇ Given $X \subset G$, $\langle X \rangle := \bigcap \{S : S \leq G \text{ and } X \subset S\}$ is the subgroup of G generated by X. If $|X| = n < \infty$, we write $\langle X \rangle = \langle a_1, a_2, \ldots, a_n \rangle$.

Note. If $H, K \leq G$, then we denote $\langle H \cup K \rangle$ as $H \vee K$.

◇ If $\varnothing \neq X \subset G$, then a *word* on X is an expression of the form $x_1^{e_1} x_2^{e_2} \ldots x_n^{e_n} = w \in G$, where $x_j \in X$ and $e_j = \pm 1$.

■ **2.7.** $\langle X \rangle \leq G$ is precisely the set of all words on X. $\to$ Let W the set of all words, prove that $W \leq G$ is the smallest : $X \subset W$

Lagrange's Theorem

◇ Let $S \leq G$ and $t \in G$. The *right coset* of S in G is the subset $St := \{st : s \in S\}$, we call t a *representative* of St (similarly we define tS).

Example $\langle n \rangle =: S < \mathbb{Z}$. Given $a \in \mathbb{Z},\ a + S = \{a + qn : q \in \mathbb{Z}\} = [a] \in \mathbb{Z}_n$.

Note. In general tS and St need not coincide, may be different.

□ **2.8.** $Sa = Sb$ (or $aS = bS$) $\leftrightharpoons\ ab^{-1} \in S$ (or $b^{-1}a \in S$).

($\Rightarrow$) $\mathbb{1}a \in Sa = Sb$, so $\exists s \in S : a = sb \ \leftrightharpoons\ ab^{-1} = s \in S$. ($\Leftarrow$) Prove two inclusions using $Sa \ni x = sa = s\sigma b \in Sb : \sigma = ab^{-1}$ (similarly for the other).

■ **2.9.** For any two right (or two left) cosets of S in G are either identical or disjoint.
$\to$ If $x = sa = tb$, then $ab^{-1} = s^{-1}t \in S$, result follows from ■ 2.8.

■ **2.10.** The number of right cosets is equal to the number of left cosets of S in G.

Proof that $f : \mathscr{L} \to \mathscr{R}$ such that $Sa \mapsto a^{-1}S$ is a well defined bijection.

◇ The *index* of S in G, denoted by $[G : S]$, is the number of right cosets.

■ **2.11.** (***Lagrange***) If $|G| < \infty$ and $S \leq G$, then $|S|$ divides $|G|$ and $[G : S] = {}^{|G|}/_{|S|}$.

By ■ 2.9. $G = St_1 \sqcup St_2 \sqcup \ldots \sqcup St_n$, and so $|G| = \sum |St_j|$, how $f_j : S \mapsto St_j$ is bijection, $|St_j| = |S|$ and $|G| = n|S|$, where $n = [G : S]$.

⊠ **2.12.** If $|G| < \infty$ and $a \in G$, then the order of a divides $|G|$. Moreover, if $|G| = p$ prime, then $|G|$ is a cyclic group. $\to$ Takes $a \neq \mathbb{1}$, then $|\langle a \rangle|$ divides $p \leftrightharpoons |\langle a \rangle| = p$

⊠ **2.14.** (***Fermat***) If p is prime and $a \in \mathbb{Z}$, then $a^p \equiv a \pmod p$.

Let $\mathbb{F} = \mathbb{Z}_p$ and $G = (\mathbb{F}^\times, \cdot)$. If $[a] = [0]$, then clearly $[a]^p = [a] = [0]$. Otherwise, $[a]^{p-1} = [\mathbb{1}]$, so $[a]^p = [a]$, any case $a^p \equiv a \pmod p$.

Cyclic Groups

□ **2.15.** If $G = \langle a \rangle : |G| = n < \infty$, then $\exists! S \leq G : |S| = d$ divides n.

Let $H = \langle b \rangle : |H| = d$ (Ex. 18), then $b^d = \mathbb{1} = (a^m)^d$, so $\exists k \in \mathbb{N} : md = nk$ (Ex. 13). Finally, $b = a^m = (a^{n/d})^k$ and $H \leq \langle a^{n/d} \rangle =: S \leq G$ (Ex. 11). Since $|S| = d$, we have $H = S$.

◇ $\varphi(n) := \#\{k \in \mathbb{N} : k < n \text{ and } (k, n) = 1\}$.

■ **2.16.** If n is a positive integer, then $n = \sum_{d \mid n} \varphi(d)$.

Let $G = \langle a \rangle : |G| = n$, then $G = \bigsqcup \operatorname{gen} C : C \leq G$. By □ 2.15. $\forall d \in \operatorname{Div}(n)$, $\exists! C_d \leq G$, so $|G| = \sum_{d \mid n} |\operatorname{gen} C_d| = \sum_{d \mid n} \varphi(d)$ (Ex. 20).

■ **2.17.** $G : |G| = n$ is cyclic $\leftrightharpoons \forall d : d \mid n,\ \exists! C_d \leq G : |C_d| = d$.

($\Rightarrow$) □ 2.15.. ($\Leftarrow$) By ■ 2.16. $n = \sum |\operatorname{gen} C| \leq \sum \varphi(d) = n$, so G must have a subgroup of order $n \leftrightharpoons G$ is cyclic.

■ **2.18.** $G \leq \mathbb{F}^\times : |G| < \infty$ is cyclic. In particular, $\mathbb{F}^\times$ is cyclic if $|\mathbb{F}| < \infty$. $\rightarrow$ The polynomial $x^d - \mathbb{1}$ has almost d roots in $\mathbb{F}[x]$, result follows from ■ 2.17.

■ **2.19.** Let p prime. A group $G : |G| = p^n$ is cyclic $\leftrightharpoons$ it's an abelian group and $\exists! S \leq G : |S| = p$. $\rightarrow$ Look [1, pag. 29], needs (Ex. 13)

Normal Subgroups

$$\diamond \quad ST := \{st : s \in S,\ t \in T\} \subset G.$$

■ **2.20.** (***Product Formula***) If $S, T \leq G : |G| < \infty$, then $|ST||S \cap T| = |S||T|$. [1] $\rightarrow$ Doesn't support sumaries, look if you want [1, pag. 30]

$\diamond$ $K \leq G$ is a *normal subgroup*, denoted by $K \lhd G$, if $\forall g \in G,\ gKg^{-1} = K$.

Example If $f : G \to H$ is homomorphism, then $\ker f \lhd G$.

Note. $K \lhd G \leftrightharpoons Kg = gK$, so there's a *partial commutativity*, if $g \in G$ and $k \in K$, $\exists k' \in K : gk = k'g$.

$\diamond$ A *conjugate* of $x \in G$ is an elemente of the form $axa^{-1} : a \in G$.

■ **2.21.** If $N \lhd G$, then the cosets form a group, the *quotient group*, denoted by G/N, of order $[G : N]$. $\rightarrow$ Set $Na * Nb := NaNb$ and prove the properties

⊠ **2.22.** If $N \lhd G$, then the *natural map*, $\pi : G \to G/N$ such that $a \mapsto Na$, is a surjective homomorphism with $\ker \pi = N$. $\rightarrow$ It's clear

Note. The quotient is the generalization of the construction of $\mathbb{Z}_n$ as $\mathbb{Z}/\langle n \rangle$ and all normal subgroup is kernel of some homomorphism.

$\diamond$ Given $a, b \in G$, we denote the *commutator* as $[a, b] := aba^{-1}b^{-1}$ and the *commutator subgroup* as G', the generated by all the commutators.

Example $X := \{[a, b] : a, b \in G\}$ need not to be a subgroup as $[a, b] * [c, d]$ need not to be a commutator (Ex. 43).

■ **2.23.** $G' \lhd G$. Moreover, if $H \lhd G$, then G/H is abelian $\leftrightharpoons G' \leq H$.

[1] ST need not to be a subgroup.

If $f : G \to G$ is homomorphism, then $f([a,b]) = [f(a), f(b)]$ so $G' \lhd G$ (Ex. 34); Moreover, $Hab = Hba \leftrightharpoons (ab)(ba)^{-1} = aba^{-1}b^{-1} = [a,b] \in H$.

The Isomorphism Theorems

References

[1] Rotman, J. J. (1984). *An introduction to the theory of groups*. Boston: Allyn and Bacon.